

Unit 7, Lesson 11: Multiplying Integers Using Models

Benchmark

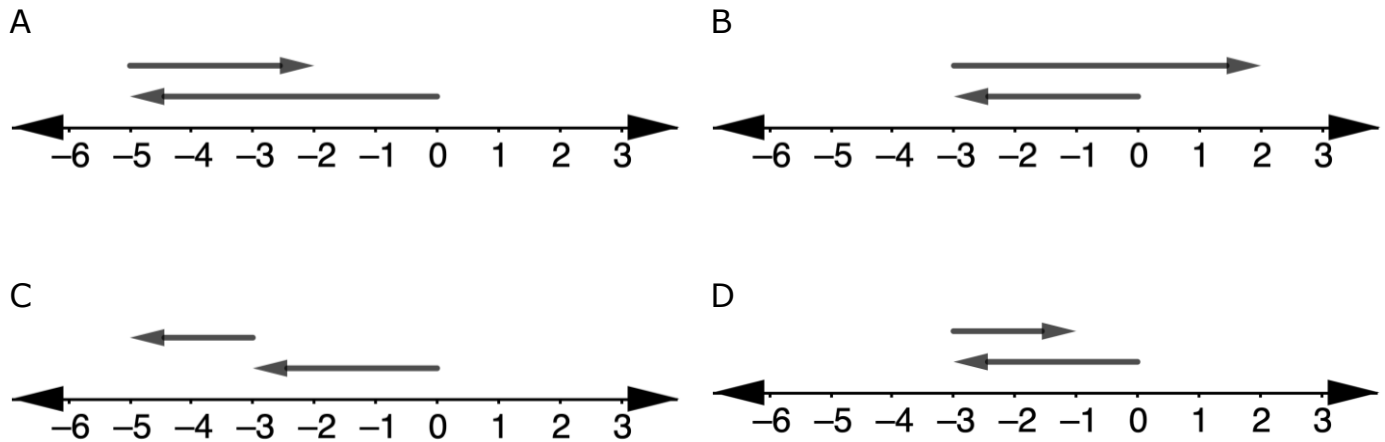
MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Learning Target

- ☐ I can multiply integers using integer chips, number lines, and patterns.

7.11.1 Warm-Up: Which One Doesn't Belong?

1. Four number lines modeling addition are shown.

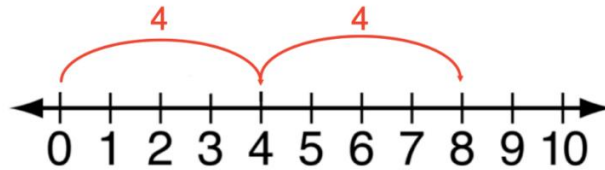


Explain which one doesn't belong.

7.11.2 Guided Instruction: Multiplying Integers Using the Number Line

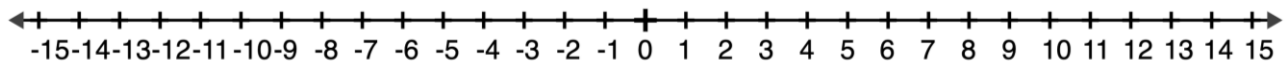
Recall that $(2)(4)$ means 2 groups with 4 units in each group. The product is the total number of objects.

This can be represented on the number line as shown.



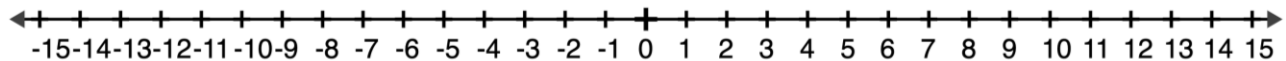
1. Represent each multiplication expression on the number line. Then, use it to complete the equation.

$$(2)(-4)$$



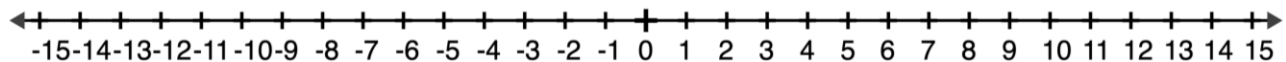
$$(2)(-4) = \underline{\hspace{2cm}}$$

$$(-2)(4)$$



$$(-2)(4) = \underline{\hspace{2cm}}$$

$$(-2)(-4)$$



$$(-2)(-4) = \underline{\hspace{2cm}}$$

2. Verify that the rules for multiplication you created using the integer chips match the results of multiplying with the number line. Make revisions to your rules if needed.

7.11.3 Guided Practice: Multiplying Integers Using Models

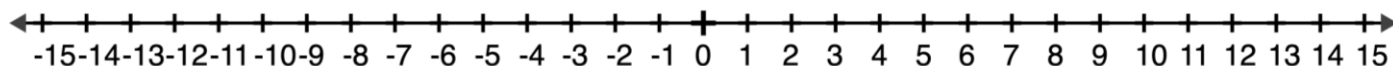
1. Use integer chips to model and evaluate each expression.

a. $(3)(-5) =$

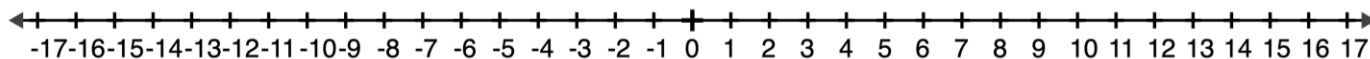
b. $-2 \times -2 =$

2. Use a number line model to evaluate each expression.

a. $(8)(-1) =$



b. $(-4)(-4) =$



3. Which model for multiplication of integers do you prefer: multiplication chips or the number line?

7.11.4 Your Turn!

1. Find the value of each expression using your preferred method.

a. 2×-8

b. $-6 \cdot (-3)$

c. $(-7)(-6)$

d. $5(-3)$

e. -9×-4

f. $7 \cdot 5$

7.11.5 **Wrap-Up: Ticket Out the Door**

1. Find the value of each expression using your preferred method.

Expression	Model	Value
$-3 \cdot -3$		
-2×4		
$(8)(-2)$		
$(3)(-8)$		